Definition: The *simple random walk* on $\mathbb{Z}^d$ is given by a walker starting at $(0,\dots,0) \in \mathbb{Z}^d$, then at time t, moves from position

$$x(t) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

to position

$$x(t+1) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

with probability $1/(2d)$ for all $1 \le i \le d$ and every discrete time unit.

Our question is somewhat simple: when, if ever, does the walker return to the origin? This is the substance of Polya's Theorem and its generalization, Varopoulos's Theorem.

The first job is to formalize the setting for this question.

Markov Chains

Setup and Basic Definition

Definition: Let X be a finite or countable set. Let $P = (p(x,y))_{x,y\in X}$ be a *stochastic matrix*, a matrix satisfying $p(x,y) \ge 0$ for all $x,y \in X$ and

$$\sum_{y\in X} p(x,y) = 1$$

for all $x \in X$.

We call P the one-step transition probability matrix. The *initial distribution* is a map $\pi\colon X \to \mathbb{R}$ such that $\pi(x) \ge 0$ for all $x \in X$ and $\sum_{x\in X}\pi(x) = 1$. The triple (X,P,π) is called a *Markov chain*.

We may formalize this as follows. Let

$$\Omega = X^{\mathbb{N}}$$

be the space of trajectories. For a fixed $y_0,\dots,y_m \in X$, we let

$$C(y_0,\dots,y_m) = \bigcap_{j=0}^{m} Z_j^{-1}\left(\{y_j\}\right)$$

be the cylinder set with base tuple $(y_0,\dots,y_m) \in X^{m+1}$, where we let $Z_j\colon \Omega \to X$ be the projection map. Define a probability measure $\mathbb{P}_\pi$ by

$$\mathbb{P}_\pi(C(y_0,\dots,y_m)) = \pi(y_0)p(y_0,y_1)\cdots p(y_{m-1}y_m).$$

By the Kolmogorov Extension Theorem, we may uniquely extend $\mathbb{P}_\pi$ to the σ-algebra $\mathcal{B} \subseteq P(\Omega)$ generated by the cylinder sets. We will let $\mathbb{P}$ be this probability measure.

The projection maps Z_j are then X-valued random variables reflecting the position of the walker at time j. The projection maps have the special property known as memorylessness — that is,

$$\begin{aligned} p(x,y) &= \mathbb{P}(Z_{n+1} = y | Z_n = x) \\ &= \mathbb{P}(Z_{n+1} = y | Z_n = x \cap Z_{n-1} = x_{n-1} \cap \cdots \cap Z_0 = x_0). \end{aligned}$$

This is also known as the Markov property.

Conversely, it can be shown that if $(Z_n)_n$ is a sequence of X-valued random variables satisfying the Markov property, then there is a Markov chain (X,P,π) that can be found by taking

$$\pi(x) = \mathbb{P}(Z_0 = x)$$

and determining p as follows: if there exists $n \in \mathbb{N}$ such that $P(Z_n = x) > 0$, set $n(x)$ to be the minimal such value, and else set $n(x) = \infty$; then, set

$$p(x,y) = \begin{cases} P\Big(Z_{n(x)+1} = y | P\Big(Z_{n(x)=x}\Big)\Big) & n(x) < \infty \\ \Phi(y) & \text{else} \end{cases}$$

for all $x, y \in X$, where $\Phi \colon X \to [0,1]$ is a map such that $\sum_{y \in X} \Phi(y) = 1$. If $n(x) < \infty$ for all $x \in X$, then this Markov chain is unique.

Note that Φ is, in a sense, an atomic probability measure.

Definition: Given a Markov chain (X, P), we let $\mathcal{G} = \mathcal{G}(X,P) = (X,E)$ be the directed graph associated with (X,P) defined as follows. The vertex set is the state space of the Markov chain, and $E \subseteq X \times X$ is the set of pairs $(x,y) \in X \times X$ such that $p(x,y) \neq 0$.

For any $n \in \mathbb{N}$, we set $P^0 = I = \left(\delta_{x,y}\right)_{x,y \in X}$, and letting

$$\begin{aligned} P^n &= PP^{n-1} \\ &= \left(p^{(n)}(x,y)\right)_{x,y \in X} \end{aligned}$$

be the nth power of the transition matrix P. Notice that $p^{(1)}(x,y) = p(x,y)$, and

$$p^{(n+1)}(x,y) = \sum_{u \in X} p(x,u) p^{(n)}(u,y).$$

Provided $\pi(x) > 0$, one has that $p^{(n)}(x,y) = \mathbb{P}(Z_n = y | Z_0 = x)$. That is, $p^{(n)}(x,y)$ gives the probability that starting from state x one arrives in state y in n random steps. In particular, we have that $\mathbb{P}(Z_{k+n} = y | Z_k = x) = p^{(n)}(x,y)$.

Suppose that $\pi = \delta_{x_0}$ is the initial configuration, specifically expressing the point mass at the starting point. Let $v_n^x(\omega) = \delta_x(Z_n(\omega))$. Then, v_n^x equals 1 if at time n, the random walker with trajectory ω is exactly at x, and 0 otherwise. Then,

$$\begin{aligned} E[v_n^x] &= \int v_n^x(\omega)\, d\mathbb{P}_{x_0}(\omega) \\ &= \mathbb{P}_{x_0}\Big(\bigcup \{C(x_0, y_1, \ldots, y_{n-1}, x) : y_1, \ldots, y_{n-1} \in X\}\Big) \\ &= \sum_{y_1, \ldots, y_{n-1} \in X} \mathbb{P}_{x_0}(C(x_0, y_1, \ldots, y_{n-1}, x)) \\ &= \sum_{y_1, \ldots, y_{n-1} \in X} p(x_0, y_1) p(y_1, y_2) \cdots p(y_{n-1}, x) \qquad \text{(Definition of } \mathbb{P}_{x_0}.) \\ &= p^{(n)}(x_0, x). \end{aligned}$$

Thus, we have that

$$v^x = \sum_{n=0}^{\infty} v_n^x$$

represents the total number of times the random walker starting at x_0 visits x. The corresponding expected value is given by

$$E[v^x] = \sum_{n=0}^{\infty} p^{(n)}(x_0, x).$$

Irreducibility

Definition: A Markov chain (X, P, π) is called *irreducible* if for every $x, y \in X$, there exists some $n \in \mathbb{N} \setminus \{0\}$ such that $p^{(n)}(x, y) > 0$.

Equivalently, the associated graph for the Markov chain is connected. We call the power series

$$G(x, y|z) = \sum_{n=0}^{\infty} p^{(n)}(x, y) z^n$$

the *Green's function* for the Markov chain.

Proposition: Suppose (X, P, π) is irreducible, and $0 < t < \infty$. Then, either $G(x, y|t)$ converges for every $x, y \in X$ or diverges for every $x, y \in X$.

Proof. Let $x, y, x', y' \in X$, and suppose that $G(x, y|t) < \infty$. We will show that $G(x', y'|t) < \infty$.

By irreducibility, we may find $h, k \in \mathbb{N} \setminus \{0\}$ such that $p^{(h)}(x, x') > 0$ and $p^{(k)}(y', y) > 0$. Furthermore,

$$p^{(h+n+k)}(x, y) \geq p^{(h)}(x, x') p^{(n)}(x', y') p^{(k)}(y', y)$$

for all $n \in \mathbb{N}$. Thus,

$$\begin{aligned}
G(x, y|t) &= \sum_{m=0}^{\infty} p^{(m)}(x, y) t^m \\
&\geq \sum_{m=h+k}^{\infty} p^{(m)}(x, y) t^m \\
&= \sum_{n=0}^{\infty} p^{(h+k+n)}(x, y) t^{h+k+n} \\
&\geq \sum_{n=0}^{\infty} t^{h+k} t^n p^{(h)}(x, x') p^{(n)}(x', y') p^{(k)}(y', y) \\
&= p^{(h)}(x, x') p^{(k)}(y', y) t^{h+k} \sum_{n=0}^{\infty} p^{(n)}(x', y') t^n \\
&= p^{(h)}(x, x') p^{(k)}(y', y) t^{h+k} G(x', y'|t),
\end{aligned}$$

whence $G(x', y'|t) < \infty$. □

We will always assume that our Markov chains are irreducible. Set

$$\begin{aligned}
G(x, y) &= G(x, y|1) \\
&= \sum_{n=0}^{\infty} p^{(n)}(x, y) \\
&\in [0, \infty].
\end{aligned}$$

Given $x, y \in X$, the real random variable

$$T^x = \inf\{n \geq 1 : Z_n = x\}$$

is called a hitting time. This is the first time that the random walker with trajectory ω visits the state x. The quantity

$$f^{(n)}(x, y) = \mathbb{P}_x(T^y = n)$$

gives the probability that the random walker, after visiting x, arrives in y for the first time in n steps. This is called the hitting probability. Notice that $f^{(0)}(x, y) = 0$ for all $x, y \in X$. Denote by

$$F(x, y|z) = \sum_{n=1}^{\infty} f^{(n)}(x, y) z^n$$

as the associated generating function.

Proposition: Suppose the Markov chain (X, P, π) is irreducible. Let $x, y \in X$ be distinct elements. Then, the following hold:

(i) $G(x,y|z) = G(y,y|z)F(x,y|z)$;

(ii) $G(x,x|z) = 1 + G(x,x|z)F(x,x|z)$;

(iii) if $0 \leq t < \infty$ is such that $G(x,x|t) < \infty$ and $F(x,x|t) < 1$, then

$$G(x,x|t) = \frac{1}{1 - F(x,x|t)}.$$

Proof. Let $n \geq 1$, $Z_0 = x$, and suppose $Z_n = y$. Denote by k the value such that $T^y = k$. We have the equality

$$\{T^y = k\} = \{Z_k = y\} \cap \Big\{Z_j \neq y \text{ for all } j = 1, 2, \ldots, k-1\Big\}.$$

Using the Markov property, we may write

$$\begin{aligned}
p^{(n)}(x,y) &= \mathbb{P}_x(Z_n = y) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y \cap T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y | T^y = k)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x\Big(Z_n = y | Z_k = y \cap Z_j \neq y \text{ for all } j = 1, \ldots, k-1\Big)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y | Z_k = y)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y).
\end{aligned}$$

Thus, we have

$$p^{(n)}(x,y) = \sum_{k=0}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y),$$

so that

$$\begin{aligned}
G(x,y|z) &= \sum_{n=0}^{\infty} p^{(n)}(x,y) z^n \\
&= p^{(0)}(x,y) + \sum_{n=1}^{\infty} p^{(n)}(x,y) z^n \\
&= \delta_x(y) + \sum_{n=1}^{\infty} \sum_{k=0}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y) z^n \\
&= \delta_x(y) + G(y,y|z)F(x,y|z).
\end{aligned}$$

Now, if $0 \leq t < \infty$ is such that $G(x,x|t) < \infty$ and $F(x,x|t) < 1$, then by a theorem of Mertens, we have that the series $G(x,x|t)F(x,x|t)$ converges, so that we get our desired result upon dividing. □

Recurrence and Transience

Definition: Let (X, P, π) be a Markov chain. A state $x \in X$ is said to be *recurrent* if the probability that the Markov chain returns to x is 1. Else, we say it is transient.

In symbols, x is recurrent if and only if

$$\mathbb{P}_x(Z_n = x \text{ for some } n \in \mathbb{N}) = \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = x\}\right) = 1.$$

Equivalently, x is recurrent if and only if $P(T^x < \infty) = 1$.

For any $x, y \in X$, let

$$\begin{aligned} F(x,y) &= \mathbb{P}_x(Z_n = y \text{ for some } n \in \mathbb{N}) \\ &= \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = y\}\right) \\ H(x,y) &= \mathbb{P}_x(Z_n = y \text{ infinitely often}) \\ &= \mathbb{P}_x\left(\limsup_{n\to\infty}\{Z_n = y\}\right). \end{aligned}$$

A state $x \in X$ is recurrent if and only if $F(x,x) = 1$. Notice that

$$\begin{aligned} F(x,y) &= \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = y\}\right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}_x(T^y = n) \\ &= \sum_{n=0}^{\infty} f^{(n)}(x,y) \\ &= F(x,y|1). \end{aligned}$$

References

[CSD21] Tullio Ceccherini-Silberstein and Michele D'Adderio. *Topics in Groups and Geometry — Growth, Amenability, and Random Walks*. Springer Monographs in Mathematics. With a foreword by Efim Zelmanov. Springer, Cham, [2021] ©2021, pp. xix+464. ISBN: 978-3-030-88108-5; 978-3-030-88109-2. DOI: 10.1007/978-3-030-88109-2. URL: https://doi.org/10.1007/978-3-030-88109-2.